

Worksheet — Assignment 2

Course: COSC2671— Social Media and Network Analysis

Project Title: Football Fan Community Analysis During the FIFA World Cup 2022

Group Members: s4123717, s4127260

Repository Platform: GitHub

1. Team Project Plan

The objective of the project was to analyse online football fan communities during the FIFA World Cup 2022 using Natural Language Processing (NLP) and Social Network Analysis (SNA) techniques. The project aimed to identify influential users, detect communities, analyse discussion sentiment, and explore dominant discussion topics within Reddit football discussions.

Week	Planned Tasks	Outcome	Status
Week 7	Topic selection, research question development, dataset investigation, repository setup.	Finalised football Reddit analysis topic and created GitHub repository.	Completed
Week 8	Collect Reddit data using Arctic Shift, perform preprocessing, organise project structure.	Collected Reddit posts/comments dataset and cleaned data for analysis.	Completed
Week 9	Perform exploratory data analysis, sentiment analysis, and topic modelling.	Generated sentiment metrics, topic clusters, and initial visualisations.	Completed
Week 10	Construct interaction network, compute centrality metrics, detect communities.	Created directed weighted graph and performed PageRank/Louvain analysis.	Completed
Week 11	Prepare final report, clean notebooks, organise repository, prepare presentation.	Finalised report, visuals, worksheet, and submission materials.	Completed

2. Weekly Timesheets

Timesheet — s4123717

Week	Tasks Completed	Hours
Week 7	Topic selection, literature review, repository setup	4
Week 8	Data extraction and preprocessing	6
Week 9	EDA, sentiment analysis, topic modelling	6
Week 10	Network analysis, centrality, community detection	5
Week 11	Report refinement, GitHub management, presentation preparation	5

Timesheet — s4127260

Week	Tasks Completed	Hours
Week 7	Topic selection, literature review, repository setup	4
Week 8	Data extraction and preprocessing	6
Week 9	EDA, sentiment analysis, topic modelling	6
Week 10	Network analysis, centrality, community detection	5
Week 11	Report refinement, GitHub management, presentation preparation	5

3. Contribution Summary

Team Member	Primary Contributions	Contribution %
s4123717	GitHub repository management, Reddit data collection, preprocessing, notebook organisation, network analysis, PageRank analysis, report writing, environment configuration, submission preparation.	50%
s4127260	Exploratory data analysis, visualisation development, sentiment analysis, topic modelling, report refinement, presentation preparation, interpretation of findings.	50%

4. Individual Self Reflection

Reflection — s4123717

During this project, I contributed primarily to GitHub repository management, dataset collection, preprocessing, network analysis, notebook organisation, and report development. One of the most challenging aspects of the project was collecting large-scale Reddit data without direct API access. This required researching alternative solutions such as Arctic Shift and creating custom extraction scripts to automate data collection.

Another challenge involved configuring a stable Python environment for topic modelling libraries such as gensim and scipy. Resolving dependency conflicts improved my understanding of Python virtual environments and reproducible workflows.

The project significantly improved my understanding of Social Network Analysis and Natural Language Processing. I gained practical experience applying PageRank, betweenness centrality, Louvain community detection, sentiment analysis, and topic modelling to a real-world social media dataset.

One aspect that worked particularly well was integrating NLP and network analysis into a single analytical pipeline. Combining sentiment trends, topic clusters, and influence metrics produced more meaningful insights than using isolated techniques.

If repeating the project, I would begin report writing earlier and explore transformer-based NLP approaches to better capture sarcasm and football-specific language.

Reflection — s4127260

Throughout the project, I contributed to exploratory data analysis, sentiment analysis, visualisation development, topic modelling, report refinement, and presentation preparation. A major challenge involved processing and analysing large-scale Reddit discussions while ensuring that visualisations remained clear and interpretable.

The project improved my technical skills in data preprocessing, visual communication, and social media analytics. I gained experience using Python libraries such as pandas, matplotlib, NetworkX, and gensim to analyse online football communities during the FIFA World Cup.

The most interesting aspect of the project was analysing how emotional events during the tournament influenced discussion sentiment and interaction patterns between users. Community detection and centrality analysis provided useful insights into how influential users shaped discussions.

If improving the project further, I would explore comparisons across multiple football-related subreddits and investigate more advanced machine learning methods for sentiment classification and topic extraction.

5. Repository and Workflow Notes

The project repository was managed using GitHub for collaboration and version control. The notebooks directory contains the complete analysis pipeline, including exploratory data analysis, sentiment analysis, topic modelling, and network analysis.

Although several Python script files exist in the scripts folder, the primary analysis workflow was performed directly within Jupyter notebooks to support iterative experimentation, visualisation, and step-by-step analysis. The scripts folder mainly contains data collection utilities and placeholder files used during project structuring.